

ADAM DSOUZA

adamdsouza0078@gmail.com | (480) 742-2651 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Full-stack engineer with 3 years of production experience across cloud-native AI/ML systems, distributed backends, and infrastructure — building systems that connect technical architecture to measurable business outcomes.

Professional Experience

Synechron | *Software Engineer* | NYC, NY

Mar '24 – Present

- Engineered a production-grade OCR pipeline (Python, OpenCV, Tesseract) to extract structured data from unstructured client documents at scale, increasing digitization throughput by 40% and eliminating manual bottlenecks blocking downstream workflows.
- Delivered a high-priority client demo under a one-week deadline by resolving multiple React frontend tickets integrated with Python and Node.js backend services — personally owning the frontend layer while coordinating planning, environment readiness, and cross-functional execution to ship core functionality on time.
- Spearheaded Java → Kotlin modernization across a production codebase, cutting build time by 25% and maintainability overhead by 20% — accelerating feature velocity and reducing onboarding friction.
- Owned end-to-end frontend delivery for a large-scale enterprise client — directly interfacing with stakeholders to scope and prioritize 350+ feature requests and defect resolutions, building production React components independently, and enforcing code review and QA standards to sustain quality across client-facing web properties.

Etherea | *Frontend Engineering Intern* | Remote

Jan '23 - Dec '23

- Launched a cross-platform React Native app with a Firebase/Firestore real-time sync engine and OAuth integration, scaling to 200K+ concurrent sessions across iOS and Android and driving 40% engagement growth through A/B testing.
- Led a 2-engineer team, enforcing code review workflows and standardizing CI/CD and environment setup — reducing production bug frequency by 35% and developer onboarding time by 45%.

Projects

Sentinel — ML Inference Pipeline | GCP, Pub/Sub, BigQuery, Terraform, FastAPI, scikit-learn

- Designed a serverless, event-driven ML inference pipeline on GCP — streaming live market data through Pub/Sub into Cloud Run workers for real-time feature extraction (RSI, MACD, Bollinger Bands, ATR), feeding a scikit-learn classifier that produces BULLISH/BEARISH/NEUTRAL signals with confidence scores at sub-100ms latency.
- Provisioned full GCP infrastructure with Terraform via GitHub Actions CI/CD; served predictions via FastAPI, persisted signals to BigQuery for downstream analytics, and managed session state in Firestore — sustaining sub-100ms p99 response times under load.

Lens — Semantic Code Search Engine | Python, FastAPI, Qdrant, tree-sitter, OpenAI, Cloud Run

- Built a semantic code search engine that indexes any GitHub repo in plain English — parsing code at function/class level via tree-sitter AST analysis, embedding chunks, and performing sub-second similarity search via Qdrant — cutting code discovery and onboarding time across large codebases.
- Developed a pluggable embedding pipeline supporting jina-embeddings-v2 (self-hosted), OpenAI text-embedding-3-small, and Ollama (local) — giving teams control over cost, privacy, and latency; deployed on GCP Cloud Run with a React UI returning file path, line numbers, symbol name, and relevance score.

Prism — API Gateway & Rate Limiting | Go, JWT, WebSockets, React, Docker, Cloud Run

- Architected a production-grade API gateway in Go with JWT authentication middleware, per-client sliding-window rate limiting, and reverse proxy routing — enabling API monetization, fine-grained access governance, and per-client billing visibility.
- Shipped a WebSocket-powered admin dashboard streaming real-time traffic, quota, and auth metrics to a React front end; packaged with Docker and deployed to GCP Cloud Run with health-check-gated rolling deployments.

CodeCollab — Real-Time Code Editor | Node.js, React, Y.js, WebSockets, Redis, PostgreSQL, Docker

- Orchestrated a real-time collaborative code editor supporting 10+ languages at <25ms sync latency — leveraging Y.js CRDT via WebSocket and Monaco Editor for conflict-free multi-user collaboration with passcode-protected sessions.
- Deployed a Dockerized Node.js backend with Redis session caching and PostgreSQL persistence for 10K+ code snippets; supported 100+ concurrent users in stress tests with sandboxed multi-language code execution via Wandbox.

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, Kotlin, Java, SQL, Bash

Frontend: React, React Native, Next.js, Redux, Tailwind CSS, Material-UI

Backend: Node.js, Express, Spring Boot, FastAPI, GraphQL, REST APIs

Cloud & DevOps: GCP (Cloud Run, Pub/Sub, BigQuery, Firestore), AWS (Lambda, EC2, S3, RDS), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins)

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Firebase

AI/ML: scikit-learn, Qdrant, sentence-transformers, OpenAI API, RAG, LLM integration, vector embeddings

Education

Bachelors in Computer Science - Arizona State University, Tempe, AZ | 3.64/4.0

Jan 2020 - Dec 2023

Honors: Dean's List (4 semesters), New American University Scholar Award